Elena Bortolato

Curriculum Vitae

Contact Information

CUNEF Universidad, Madrid, Spain
Web: elenabortolato.github.io

Current Position

Since 2026	Assistant Professor, CUNEF Universidad, Madrid

Research Interests

- Bayesian and frequentist inference, robustness, and model misspecification
- Computational statistics and Monte Carlo methods, with particular emphasis on Markov Chain Monte Carlo (MCMC) algorithms on manifolds
- Simulation-based (likelihood-free) inference, uncertainty quantification, and measures of statistical evidence (including p-values and confidence distributions), together with their interplay with probabilistic machine learning
- Latent variable models, factor analysis, and statistical methods for high-dimensional data
- Approximate inference and objective methods in Bayesian statistics
- Applications of statistical methodology to biomedicine and health sciences

Education

Oct 2020 – May 2024	**PhD in Statistical Sciences**, University of Padova, Department of Statistical Sciences Thesis: *Advancements in distribution sampling for statistical inference* Supervisor: prof. Laura Ventura, University of Padova Co-supervisors: prof. Pierre E. Jacob, ESSEC; prof. Robin J. Ryder, Imperial College London
Oct 2018 – Sep 2020	**Master degree (Laurea Magistrale) in Statistical Sciences**, University of Padova, Department of Statistical Sciences Thesis: *Bias reduction in the equicorrelated multivariate normal* Supervisor: prof. Euloge Clovis Kenne Pagui. Final mark: 110/110 cum laude
Oct 2015 – Jul 2018	**Bachelor degree (Laurea Triennale) in Statistics for Technology and Sciences**, University of Padova, Department of Statistical Sciences Thesis: *Models and methods for music analysis* Supervisor: prof. Alessandra Rosalba Brazzale. Final mark: 106/110

Work Experience

Since 2026	Assistant Professor, CUNEF Universidad, Madrid
Oct 2024 – 2026	Postdoctoral Research Fellow, Universitat Pompeu Fabra and Barcelona School of Economics Affiliated Professor, Barcelona School of Economics Supervisors: prof. David Rossell (UPF) and prof. Stephen Hansen (UCL)

Nov 2023 – Oct 2024	Research Fellow, University of Padova Project: *Bayesian nonparametric spatio-temporal processes and their applications* Supervisor: prof. Antonio Canale, University of Padova
Aug 2019 – Sep 2019	Social Media Analyst (AIESEC volunteer), ACAM Mocotó, Florianópolis, Brazil
Feb 2019 – Jul 2019	Research Fellow, University of Padova Project: *ROI-MOB: Measuring return on investment from EU VET mobility* (2016-1-IT01-KA202-005396)

Awards and Scholarships

2024	Best Poster Award, ISBA 2024, Venice
2023	Honorable mention (Poster), GRASPA 2023, Palermo
2023	ISBA/BayesComp Travel Grant
2022	Laboratoires Junior financing (PhD student teams prize, University of Lyon, University Jean Monnet)
2021	Best thesis in Mathematical Statistics in memory of professor Oliviero Lessi, Società Italiana di Statistica (SIS)
2020	First Prize, Laboratorio con le aziende (Business Lab), Department of Statistical Sciences, University of Padova and ASPIAG

Computer Skills

R, RStudio, Julia, Python, SAS, C++, LaTeX

Language Skills

Italian: native. English: fluent. French: basic. Spanish: basic. Catalan: basic.

Publications

Articles in journals

Bortolato, E., Canale, A. (2026). Adaptive Partition Factor Analysis. *Journal of the American Statistical Association*.

Bortolato, E., Ventura, L. (2026). Box-Confidence Depth: simulation-based inference with hyper-rectangles. *Electronic Journal of Statistics*.

Bortolato, E., Bertolino, F., Musio, M., Ventura, L. (2025). Bayesian Discrepancy Measure: Higher-order and Skewed approximations. *Entropy*.

Colaone et al. (2025). Effectiveness of Telemedicine Nursing Interventions in Adult Obesity: A Systematic Review and meta-analysis. *Current Obesity Reports*, accepted.

Biz, C., et al. (2025). The Learning Curve of Reverdin-Isham and Akin percutaneous osteotomies for Hallux Valgus correction: a Bayesian approach. *Journal of Clinical Medicine*, 14 (6), 1921.

Bortolato, E., Ventura, L. (2024). Objective priors for invariant e-values in the presence of nuisance parameters. *Entropy*, 26 (1), 58.

Bortolato, E., Kenne Pagui, E.C. (2024). Bias reduction and robustness for Gaussian longitudinal data analysis. *Journal of Statistical Computation and Simulation*, 94 (2), 360–375.

Bortolato, E., Ventura, L. (2023). On approximate robust confidence distributions. *Econometrics and Statistics*. https://doi.org/10.1016/j.ecosta.2023.04.006

Biz, C., et al. (2022). Treatment of Distal Radius Fractures with Bridging External Fixator with Optional Percutaneous K-Wires: What Are the Right Indications for Patient Age, Gender, Dominant Limb and Injury Pattern? *Journal of Personalized Medicine*, 12, 1532.

Scarpazza, C., et al. (2021). Idiopathic and acquired pedophilia as two distinct disorders: an insight from neuroimaging. *Brain Imaging and Behavior*, 15, 2681–2692.

Articles in conference books of papers

Bortolato, E., Musio, M., Ventura, L. (2024). Approximate Inference for the Bayesian Discrepancy Measure for Precise Statistical Hypotheses. *Methodological and Applied Statistics and Demography III*, Springer.

Bortolato, E., Ventura, L. (2022). Confidence distributions and fusion inference for intractable likelihoods. *SIS Book of short papers*.

Bortolato, E., Kenne Pagui, E.C. (2021). Bias reduction in the equicorrelated multivariate normal. *SIS Book of short papers*.

Bortolato, E., Fabbris, L., Vivian, M. (2019). Quantity and mood of final open-ended comments on an Erasmus+ VET mobility questionnaire. *ASA Book of short papers*.

Preprints and working papers

Bortolato, E., Jacob, P.E., Ryder, R.J. (2025+). Manifold MCMC for generic sampling problems and couplings. Working paper.

Bortolato, E., Hansen, S., Rossell, D. (2025+). Birth-death graphical models. Working paper.

Posters

Bortolato, E., Hansen, S., Rossell, D. Birth death graphical models. CMStatistics, London, UK, December 2025 (invited).

Bortolato, E. MCMC. ISBA World Meeting 2024, Venice, IT, July 2024.

Bortolato, E. Flexible Multi-study Factor Analysis. BAYSM 2024, Venice, IT, June 2024.

Bortolato, E. Estimating malaria parasites age distribution: a Bayesian statistical model. GRASPA 2023, Palermo, IT, July 2023.

Bortolato, E. Couplings of MCMC algorithms on manifolds. Approximate methods in Statistics, Marseille, FR, June 2023.

Bortolato, E. Manifold MCMC methods for generic target distributions.

Bortolato, E. Alternatives for approximation in likelihood-free inference: Box-ABC. ISBA 2022 World Meeting, Montreal, CA, July 2022.

Conference Presentations

Bortolato, E. Birth-death graphical models. CMStatistics, London, December 2025 (invited).

Bortolato, E. Bayesian measure of evidence. ISI, The Hague, October 2025 (invited).

Bortolato, E. Sampling on constrained spaces. SISBayes25, September 2025 (invited).

Bortolato, E. Measuring confidence with hyper-rectangles. Workshop in honour of Oliviero Lessi, May 2025 (invited).

Bortolato, E. MCMC on manifolds: challenges and solutions. Florence, Italy, May 2024 (invited).

Bortolato, E. MCMC on manifolds: challenges and solutions. Venice, Italy, March 2024 (invited).

Bortolato, E., Jacob, P.E., Ryder, R.J. Coupling MCMC algorithms on submanifolds. Bayes@CIRM, France, November 2023 (invited).

Bortolato, E., Jacob, P.E., Ryder, R.J. Tuning refinements and parallelizable computation for MCMC on manifolds. AppliBUGS 2023, Paris, France, June 2023 (invited).

Bortolato, E., Jacob, P.E., Ryder, R.J. Convergence of MCMC algorithms on manifolds through coupling techniques. BayesComp 2023, Levi, Finland, March 2023 (contributed).

Bortolato, E., Ventura, L. Box-ABC for likelihood-free inference. CMStatistics, London, UK, 17–19 December 2022 (invited).

Bortolato, E., Ventura, L. Confidence distributions and fusion inference for intractable likelihoods. Fusion workshop, Marseille, FR, October 2022 (invited).

Bortolato, E., Ventura, L. Confidence distributions and fusion inference for intractable likelihoods. Italian Statistical Society meeting, Caserta, IT, June 2022 (contributed).

Bortolato, E., Kenne Pagui, E.C. Bias reduction in the equicorrelated multivariate normal. 50th meeting of the Italian Statistical Society, Pisa, Italy, June 25, 2021 (contributed).

Teaching Experience

Main instructor

2026 – 2027	Data Analysis, CUNEF Universidad (Double degree in Business Administration and Law)
2026 – 2027	Econometrics, CUNEF Universidad (Double degree in Business Administration and Law)
Oct – Dec 2025	Computational Machine Learning, Barcelona School of Economics (Master in Data Science Methodology)
Oct – Dec 2024	Computational Machine Learning, Barcelona School of Economics (Master in Data Science Methodology)
Jan – Mar 2024	Computational Machine Learning and Deep Learning, Barcelona School of Economics (Master in Data Science for Decision Making)
Jan – Mar 2023	Business Statistics and Introduction to Data Analytics, ESSEC Business School (Grande École, Master in Management)

Teaching assistant

Nov 2024	Foundations of Data Science, Barcelona School of Economics (Master in Finance)
Feb – Jun 2024	Statistical Models 1, University of Padova (Department of Statistical Sciences, Bachelor)
Feb – Jun 2024	Mathematics and Statistics, University of Padova (Department of Biology, Bachelor)
Oct – Dec 2023	Multidimensional Data Analysis, University of Padova (Department of Statistical Sciences, Bachelor)

Tutoring

Oct 2019 – Jun 2020	Calculus (BSc in Statistics for Technology and Sciences; BSc in Statistics for Business and Economy) and Statistics (MSc in Statistical Sciences), University of Padova

Schools

Oct 2023	Bayes@CIRM, Marseille, France
Sep 2023	Sampling high-dimensional probability measures: applications in (non)equilibrium molecular dynamics and statistics, Paris, France
Sep 2023	Replicability crisis in Science?, Padova, Italy
Feb 2022	CEREMADE Winter School 2022, Saint-Pierre-Canivet, France

Visiting

Nov 2021 – Jun 2022 Nov 2022 – Jun 2023	ESSEC Business School, Cergy, France

Organisation

2026	Member of the Organising Committee, European Young Statisticians Meeting (EYSM 2026)
Jun 2023	Support to the organisation of the 14th International Conference on Monte Carlo Methods and Applications (MCM23), Paris

Referee Activity

- Bayesian Analysis (2023, 2024, 2025)
- Journal of Computational and Graphical Statistics (2025)
- Journal of Multivariate Analysis (2025)
- Journal of Statistical Computation and Simulation (2024)

11/11/2026